



Technical Service Bulletin: TSB160055

Released Date: 07-Oct-2021

SmartAccel™ Feature Release and Initial Tuning Guide

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- ISX12 CM2350 X102
- ISX15 CM2350 X101
- X15 CM2350 X114B
- X15 CM2450 X124B
- X12 CM2350 X119B
- X15 CM2450 X134B
- X12 CM2450 X137B

Manual transmissions **only**.

Issue

Customer has concerns with vehicle powertrain robustness or has experienced powertrain component malfunctions when launching the vehicle from a stop.

Resolution

The SmartAccel™ feature has been released to increase powertrain robustness during a vehicle launch. This is a user-selectable feature in INSITE™ electronic service tool. When enabled, the SmartAccel™ feature will limit the rate at which torque is applied to the flywheel. This feature is designed to protect powertrain components from impact torque from harsh events such as a clutch dump when the Accelerator Pedal Position is 100%. The SmartAccel™ feature is designed to protect powertrain components on all transmissions.

Enabling SmartAccel™ Feature

1. Determine Gear Ratio Threshold

- Connect INSITE™ electronic service tool version 8.1.2 or newer.

- Verify/Install engine control module (ECM) calibration code with SmartAccel™ feature. ECM calibration codes after 30 Mar 2016 have the SmartAccel™ feature.
- Enable SmartAccel™ in Features and Parameters.
- Determine highest gear used by customer at vehicle launch.
- Determine transmission gear ratios of the vehicle. See equipment manufacturer service information. See Table 1 below for example gear ratios.

Table 1, Example Gear Ratios

Gear Number	1	2	3	4	5	6	7	8	9	10
Gear Ratio	14.8	10.95	8.09	5.97	4.46	3.32	2.45	1.81	1.34	1.00

- Determine average gear ratio between highest launch gear and next highest sequential gear.
For Example:
 - If third gear is the highest launch gear; the average ratio of third gear and the next higher gear (fourth gear), is as follows.
 - $(8.09 + 5.97) / 2 = 7.03$
 - Set 7.03 as the gear ratio threshold for SmartAccel™.

2. Determine Torque Ramp Rate Value

- Connect INSITE™ electronic service tool version 8.1.2 or newer.
- Select Features and Parameters and select Vehicle Speed Source to see Rear Axle Ratio.
- If rear axle ratio < 2.47, then input 370 lb*ft/s [500 N•m/s] for Torque Ramp Rate.
- If rear axle ratio >= 2.47 and rear axle ratio < 3.91, then input 553 lb*ft/s [750 N•m/s] for Torque Ramp Rate.
- If rear axle ratio >= 3.91, then input 737 lb*ft/s [1000 N•m/s] for Torque Ramp Rate.

Customer Communication

Decrease in acceleration in the first few gears from launch may be observed.

Production Status

SmartAccel™ feature has been implemented for production. See Table 2.

Table 2, Production Information

Product	Engine Serial Number (ESN) First	Build Date*	Engine Plant
ISX12 CM2350 X102	75050864	6 Apr 16	Jamestown Engine Plant
ISX15 CM2350 X101	79912573	5 Apr 16	Jamestown Engine Plant

*Engine build date can be found on the engine dataplate.

Document History

Date	Details
2016-5-31	Module Created
2020-4-16	Non-Product Problem Solving (PPS)
2021-10-7	Updated Product Affected and Determine Torque Ramp Rate Value section.

Last Modified: 07-Oct-2021
